

Program Assignment 2 (Spring 2026)

Due Date: **June 02, 2026**

Express Delivery

Mary works at an express delivery company and is asked to provide delivery plans for drivers. The company expects the drivers to fulfill on delivery plan as soon as possible to expedite delivery.

A city map is a graph $G = (V, E)$ with edge weight function $w: E \rightarrow \mathcal{R}$, where weight $w(i, j)$ denotes the time traveling between vertices (cities) i and j . For example, Fig. 1 shows a city map of 7 vertices (cities). Let a path $(s, v_1, v_2, \dots, v_k, s)$ denote a *delivery plan* in G which starts from s and visits $v_1, v_2, \dots, v_k, s$ in sequence. Let the time for delivery plan $(s, v_1, v_2, \dots, v_k, s)$ be the total time of $d(s, v_1) + d(s, v_n) + \sum_{i=1}^{k-1} d(v_i, v_{i+1})$, where $d(x, y)$ denotes the minimum time travelling between x and y . The *minimum delivery plan* starting from s to destinations $v_1, v_2, \dots, v_k$ is defined as the minimum time of the plan $(s, v_{\pi(1)}, v_{\pi(2)}, \dots, v_{\pi(k)}, s)$, where $\pi: \{1, 2, \dots, k\} \rightarrow \{1, 2, \dots, k\}$ is a permutation function. For example, in Fig. 1, cycle $(1, 5, 7, 1)$ is a delivery plan starting from 1 in time $5 + 6 + 10 = 21$ and cycle $(1, 7, 5, 1)$ is also a delivery plan starting from 1 in time $10 + 6 + 5 = 21$. Therefore the minimum delivery plan starting from 1 to destinations vertices 5 and 7 is 21.

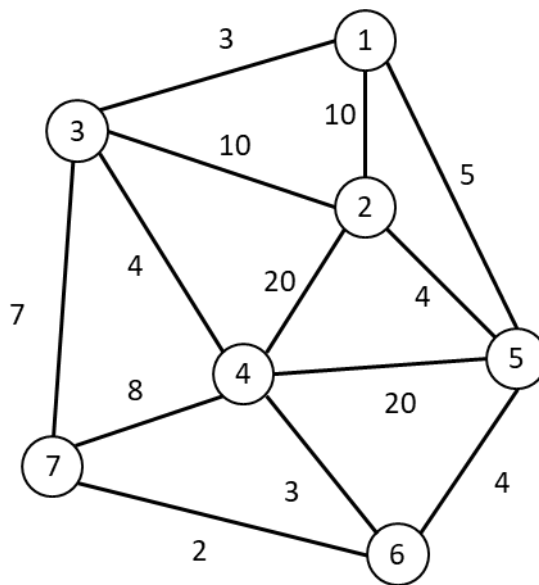


Fig. 1

Assignment Description

Please design an information system for express delivery companies to provide a minimum delivery plan from a start to several destinations. Your system design must include the following four components:

1. **Environment Description:** Begin by describing the environment of a city, including the meanings of vertices, edges and weights. Then describe the environment of an express delivery company, such as the number of vehicles and the capacity of a vehicle. In addition, the destinations vertices could be for delivery or pickup.
2. **System Design:** Propose an information system that enables the company to query a minimum delivery schedule from a start to several destinations. The plan should include the detailed delivery schedule for each vehicle. Consider user interfaces, data sources, and how updates are communicated.
3. **Problem Definition:** Clearly define the problems the system needs to address.
4. **Algorithm Design:** Propose algorithms to solve these problems. For example, shortest path algorithms. At least one algorithm must be implemented correctly, with your implementation included in the report.

Deliverables (Due Date: 13:10, June 02, 2026)

- A **hardcopy report** detailing your system design, problem definitions, and algorithm implementations.
- An **oral presentation** and **system demonstration** to introduce your work to the class.

Grading Policy

Baseline: Full credit for meeting the minimum requirements, including a clear description of the system environment, system design, problem definitions, at least one correctly implemented algorithm, and completion of both the written report and oral presentation.

Bonus: Additional credit will be awarded for:

- Defining more relevant and insightful problems in the system.
- Proposing a wider range of appropriate and creative algorithms.
- Implementing more algorithms correctly and effectively.

Grading Distribution

- Written Report: 50%
- Presentation & System Demonstration: 50%

Report Format

The report should be between **20** and **30** pages in length, formatted on **A4-size paper**. Please follow the formatting and structural guidelines below:

A. Cover Page

Include the following information clearly:

- Course title and section
- Student name
- Student ID number

B. Report Body

- Use **1.5 line spacing** throughout the report.
- Font size should be **12 point** for the main text and **14 point** for section titles.
- Each page must include a **page number**.

The body of the report should include the following sections:

1. Context, System, and Problem Definition

Describe the environment, the proposed system, and the specific problems the system aims to solve. *(It is necessary to provide a detailed breakdown of the workflow and specific interactions involved when utilizing AI tools to complete this section.)*

2. Proposed Algorithms

Detail the algorithms designed to address the defined problems, including any assumptions and design considerations. *(It is necessary to provide a detailed breakdown of the workflow and specific interactions involved when utilizing AI tools to complete this section.)*

3. Implementations and Results

Present the implemented algorithm(s), describe how they were tested, and provide the results. *(It is necessary to provide a detailed breakdown of the workflow and specific interactions involved when utilizing AI tools to complete this section.)*

4. Discussion

Analyze the strengths and limitations of your system and algorithms. Reflect on challenges encountered and possible future improvements.

Other Notes:

- Plagiarism will result in a score of zero.
- Please conduct your oral presentations in the spirit of collaborative learning, sharing, and mutual improvement. Learn from each other by exchanging insights and reflections.
- Please submit your assignments on time; **late submissions will not be accepted for credit.**